

Solutions to "Algebraic Geometry and Arithmetic Curves"

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August 31, 2026

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1 Some topics in commutative algebra

1.1 Tensor products

Exercise 1.1.1. Let $\{M_i\}_{i \in I}, \{N_j\}_{j \in J}$ be two families of modules over a ring A . Then

$$(\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j).$$

Proof. There is a chain of natural isomorphisms:

$$\begin{aligned} & (\oplus_{i \in I} M_i) \otimes_A (\oplus_{j \in J} N_j) \\ & \cong \oplus_{i \in I} (M_i \otimes_A (\oplus_{j \in J} N_j)) \\ & \cong \oplus_{j \in J} \oplus_{i \in I} (M_i \otimes_A N_j) \\ & \cong \oplus_{(i,j) \in I \times J} (M_i \otimes_A N_j). \end{aligned}$$

□

Exercise 1.1.2. There is a unique A -linear map

$$f: \text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$$

such that $f(u \otimes v) = u \otimes v$.

Proof. Consider the map $g: \text{Hom}_A(M, M') \times \text{Hom}_A(N, N') \rightarrow \text{Hom}_A(M \otimes_A N, M' \otimes_A N')$ sending (u, v) to $u \otimes v$. It is clearly bilinear, so factors uniquely through $\text{Hom}_A(M, M') \otimes_A \text{Hom}_A(N, N')$. $\square$

Exercise 1.1.3. Let M, N be A -modules and let $i: M' \rightarrow M, j: N' \rightarrow N$ be submodules of M and N , respectively. Then there is a canonical isomorphism

$$(M/M') \otimes_A (N/N') \cong (M \otimes_A N) / (\text{Im } i_N + \text{Im } j_N).$$

In particular, $(\mathbb{Z}/n\mathbb{Z}) \otimes_{\mathbb{Z}} (\mathbb{Z}/m\mathbb{Z}) = \mathbb{Z}/l\mathbb{Z}$, where $l = \gcd(m, n)$.

Proof. Observe that

$$(M \otimes_A N) / (\text{Im } i_N + \text{Im } j_N) \cong ((M \otimes_A N) / (\text{Im } i_N)) / (\text{Im } j_{M/M'}),$$

so Corollary 1.13 gives us a chain of isomorphisms

$$\begin{aligned} & ((M \otimes_A N) / (\text{Im } i_N)) / (\text{Im } j_{M/M'}) \\ & \cong ((M/M') \otimes_A N) / (\text{Im } j_{M/M'}) \\ & \cong (M/M') \otimes_A (N/N'). \end{aligned}$$

For the last statement, just recall that $(m) + (n) = (l)$. $\square$

Exercise 1.1.4. Let M, N be A -modules and let B, C be A -algebras.

- (a) If M and N are finitely generated over A , then so is $M \otimes_A N$.
- (b) If B and C are finitely generated over A , then so is $B \otimes_A C$.
- (c) The converses of (a) and (b) are in general false.

Proof. Parts (a) and (b) are obvious. For part (c), just notice that $\mathbb{Q} \otimes_{\mathbb{Z}} \mathbb{Z}/2\mathbb{Z}$ is the zero ring, but $\mathbb{Q}$ is not finitely generated over $\mathbb{Z}$. $\square$

Exercise 1.1.5. Let $\rho: A \rightarrow B$ be a ring homomorphism, M an A -module, and N a B -module. There is a canonical isomorphism of A -modules

$$\text{Hom}_A(M, \rho_* N) \cong \text{Hom}_B(\rho^* M, N).$$

Proof. Let $f: \text{Hom}_A(M, \rho_* N) \rightarrow \text{Hom}_B(\rho^* M, N)$ be the map sending u to the map $f(u)$ s.t. $f(u)(m \otimes b) = bu(m)$. This is well-defined because if $n \otimes c = m \otimes b$, we may assume that there is some $a \in A$ s.t. $an = m$ and $\rho(a)b = c$ and so $f(u)(n \otimes c) = cu(n) = \rho(a)bu(n) = bu(m)$.

In addition, define $g: \text{Hom}_B(\rho^* M, N) \rightarrow \text{Hom}_A(M, \rho_* N)$ to be the map s.t. $g(u)(m) = u(m \otimes 1)$. This is immediately seen to be well-defined, and we contest that this is an inverse for f . Indeed, $f(g(u))(m \otimes b) = bg(u)(m) = bu(m \otimes 1) = u(m \otimes b)$ and $g(f(u))(m) = f(u)(m \otimes 1) = u(m)$. $\square$

Exercise 1.1.6. Let $(N_i)_{i \in I}$ be a direct system of modules. Then for any A -module M , there exists a canonical isomorphism $\varinjlim (N_i \otimes_A M) \cong (\varinjlim N_i) \otimes_A M$.

Proof. Tensor-hom. □

Just kidding, it follows from the fact that $\varinjlim (N_i \times M)$ is canonically isomorphic to $(\varinjlim N_i) \times M$, and that a bilinear map over the latter must then induce a cocone of bilinear maps. □

Exercise 1.1.7. Let B be an A -algebra, and let M, N be B -modules. Then there exists a canonical surjection $M \otimes_A N \rightarrow M \otimes_B N$.

Proof. Do I really need to spell it out? □

1.2 Flatness

Exercise 1.2.1. Let M be an A -module and let $I \subseteq \text{Ann}(m)$ be an ideal.

- (a) There is a natural A/I -module structure on M , and $M \cong M \otimes_A A/I$.
- (b) Let N be another A -module such that $I \subseteq \text{Ann}(N)$. The canonical homomorphism $M \otimes_A N \rightarrow M \otimes_{A/I} N$ is an isomorphism.

Proof. For (a), observe that if $x \in I$, then $am = (a + x)m$ for any $a \in A$ and $m \in M$. This implies that the multiplication of elements of M by cosets $a + I$ is well defined, hence we have an A/I -module structure on M . To see that this is the extension of scalars, recall that $M \otimes_A A/I \cong M/(IM)$, which is just M .

Part (b) follows from a chain of isomorphisms

$$\begin{aligned} M \otimes_A N & \\ \cong (M \otimes_A A/I) \otimes_{A/I} N & \\ \cong M \otimes_{A/I} N, & \end{aligned}$$

where the first isomorphism comes from Corollary 1.11 and the second comes from part (a). □

Exercise 1.2.2. Let $\rho: A \rightarrow B$ be a ring homomorphism, S a multiplicative subset of A , and $T = \rho(S)$. Then T is a multiplicative subset of B , and $T^{-1}B \cong B \otimes_A S^{-1}A$.

Proof. The first part is obvious. To see the last statement, let $\iota: B \rightarrow T^{-1}B$ be the canonical embedding. Since ι sends elements of $T = \rho(S)$ to units, there is a canonical homomorphism $\phi: S^{-1}A \rightarrow T^{-1}B$. Since $\otimes_A$ is a pushout in the category of A -algebras, there is a unique homomorphism $B \otimes_A S^{-1}A \rightarrow T^{-1}B$ such that this diagram commutes:

$$\begin{array}{ccc} S^{-1}A \otimes B & \longleftarrow & S^{-1}A \\ \uparrow & \searrow & \downarrow \\ B & \longrightarrow & T^{-1}B. \end{array}$$

This is an isomorphism because of universal property of localizations. $\square$

Exercise 1.2.3. Nakayama's lemma does not hold for modules that are not finitely generated.

Proof. Consider $\mathbf{Q}$ as a module over $\mathbf{Z}_{(p)}$. $\square$

Exercise 1.2.4. Let I be an ideal of a ring A . Then the following are equivalent:

- (a) A/I is flat over A .
- (b) $I^2 = I$
- (c) There is an element $e \in I$ such that $e^2 = e$ and $eA = I$.

Proof. ((i) $\Rightarrow$ (ii)) Consider the exact sequence $0 \rightarrow I \rightarrow A \rightarrow A/I \rightarrow 0$. Since A/I is flat, we have another exact sequence $0 \rightarrow A/I \rightarrow A/I^2 \rightarrow 0$, so $I = I^2$.

((ii) $\Rightarrow$ (iii)) I don't know how to do this morally. The easy way out is to use the form of Nakayama's Lemma proven by the determinant trick.

((iii) $\Rightarrow$ (i)) Observe that in any maximal ideal, I either localizes to (0) or (1) . In any case, it's flat at every maximal ideal, hence flat overall. $\square$

1.3 Formal completion

Exercise 1.3.1. The usual topology on $\mathbf{R}$ is not given by a subgroup filtration.

Proof. Use noggin for ≥ 2 seconds. $\square$

2 General properties of schemes

2.1 Spectrum of a ring

Exercise 2.1.1. For k a field, $\text{Spec}(k[[T]])$ is the space $\{\xi, 0\}$ with generic point ξ and closed point 0 .

Proof. It is easy to see that any power series with a non-zero constant term is a unit. Furthermore, if we have $f = \sum a_n T^n$ with $a_N = 1$ for some N and $a_n = 0$ whenever $n \leq N$, then we can pull out a common factor of T^N . We see that f is then a unit multiple of T^N and $(f) = (T^N)$. Clearly, this is only prime when $N = 1$. It is easy to see that $k[[T]]$ is an entire ring, so the proposition is immediate. $\square$

Exercise 2.1.2. Let $\varphi: A \rightarrow B$ be a homomorphism of finitely generated algebras over a field k . The image of a closed point under $\text{Spec } \varphi$ is a closed point.

Proof. If $\text{Spec } \varphi$ takes the maximal ideal $\mathfrak{m}$ into the ideal $\mathfrak{p}$, we get a diagram

$$k \rightarrow A/\mathfrak{p} \xrightarrow{\tilde{\varphi}} B/\mathfrak{m}.$$

By Corollary 1.12, $B/\mathfrak{m}$ is a finite algebraic extension of k , so $B/\mathfrak{m}$ (resp. $A/\mathfrak{p}$) is finite over $A/\mathfrak{p}$ (resp. k). It then suffices to prove that $A/\mathfrak{p}$ is a field. Any $a \in A$ has an associated irreducible polynomial $P = \sum a_i X^i \in k[x]$ such that $P(a) = 0$. This polynomial must have a nonzero constant term, so we have

$$-\frac{1}{a_0} \sum_{i>0} a_i a^i = 1,$$

but the polynomial on the left has no constant term and we can pull out a factor of a . $\square$

Exercise 2.1.3. Let $k = \mathbf{R}$ be the field of real numbers and let $A = k[X, Y]/(X^2 + Y^2 + 1)$. Let x, y be the respective images of X, Y in A .

- (a) Let $\mathfrak{m}$ be a maximal ideal of A . Then there are $a, b, c, d \in k$ such that $x^2 + ax + b, y^2 + cy + d \in \mathfrak{m}$. Furthermore, $\mathfrak{m}$ is generated by a single element $f = \alpha x + \beta y + \gamma$ with at least one of α, β nonzero.
- (b) The map $(\alpha, \beta, \gamma) \mapsto (\alpha x + \beta y + \gamma)A$ establishes a bijection between the subset $\mathbf{P}(k^3) \setminus \{(0, 0, 1)\}$ of the projective space $\mathbf{P}(k^3)$ and the set of maximal ideals of A .
- (c) Let $\mathfrak{p}$ be a non-maximal prime ideal of A . Then the canonical homomorphism $\varphi: k[X] \rightarrow A$ is finite and injective, and $k[X] \cap \mathfrak{p} = 0$. Let $g \in \mathfrak{p}$ and let $g^n + a_{n-1}g^{n-1} + \dots + a_0 = 0$ be an integral equation with $a_i \in k[X]$. Then $a_0 = 0$, and so $\mathfrak{p} = 0$.

Proof. First, notice that $A/\mathfrak{m}$ is isomorphic to $\mathbf{C}$, since $X^2 + Y^2 + 1$ does not split in k . Then the existence of a, b, c, d follows from the fact that any complex number satisfies a monic quadratic equation. Since $\mathbf{C}$ is degree 2 over k , there are α, β, γ , not all zero, such that

$$\alpha x' + \beta y' + \gamma = 0,$$

where x' and y' are the respective images of x and y . Since not all of them are zero, we may assume that α is nonzero. Then we get an equation for x' in terms of y' , which we can then substitute into the equation $x'^2 + y'^2 + 1 = 0$. This new equation cannot split in k because that would imply a real root of $x'^2 + y'^2 + 1 = 0$, so we have that $\mathfrak{m} = fA$, where f is the ideal corresponding to the linear equation in x' and y' .

Part (b) holds, since a solution to $X^2 + Y^2 + 1 = 0$ is determined (up to conjugacy) by an arbitrary complex number, which itself determines a linear equation with real coefficients.

That φ is finite and injective is clear, so consider the induced homomorphism $\tilde{\varphi}: k[X]/\varphi^{-1}(\mathfrak{p}) \rightarrow A/\mathfrak{p}$. Notice that $B = k[X]/\varphi^{-1}(\mathfrak{p})$ is not a field, since $\tilde{\varphi}$ is

still finite and injective, which implies that $A/\mathfrak{p}$ is a field as well. Then $B = k[X]$ and $\mathfrak{p} \cap k[X] = 0$. Finally, in the integral equation for g , notice that a_0 must be a multiple of g , hence 0. Then the equation has a common factor of g which we may cancel, which gives a new equation with constant term a_1 . This repeats indefinitely and so $g = 0$. $\square$

Exercise 2.1.4. Let A be a ring.

- (a) Let $\mathfrak{p}$ be a minimal prime ideal of A . Then every element of $\mathfrak{p}A_{\mathfrak{p}}$ is nilpotent. In particular, every element of $\mathfrak{p}$ is a zero divisor.
- (b) If A is reduced, then any zero divisor in A is an element of a minimal prime ideal. This is not true in general if A is not reduced.

Proof. Part (a) is obvious.

For (b), let x be any zero divisor and let $\mathfrak{p}$ be a prime ideal minimal among those that contain x . If $\mathfrak{p}$ is minimal, we are done. Otherwise, suppose that $\mathfrak{q} \subset \mathfrak{p}$ is a minimal prime ideal not containing x . Since A is reduced, we must have $\sqrt{(x)} \cap \text{Ann}(x) = \{0\}$, so $\mathfrak{q}$ must contain a nonzero element $x' \in \text{Ann}(x)$. Localizing by $\mathfrak{q}$, we see that x is sent to 0 and x' is sent to a nilpotent. This implies that there is some positive integer n and some element $s \in A \setminus \mathfrak{q}$ such that $s(x'^n - x) = 0$, and it is easy to see that this means that $sx^2 = 0 \in \mathfrak{q}$, so $x^2 \in \mathfrak{q}$, contradiction. $\square$

The fact that this is false for A not reduced comes is immediate from the fact that there can be non-nilpotent zero divisors in a ring.

Exercise 2.1.5. Let k be a field. Let $\mathfrak{m}$ be a maximal ideal of $k[T_1, \dots, T_n]$.

- (a) Let $P_1(T_1)$ be a generator of $\mathfrak{m} \cap k[T_1]$ and $k_1 = k[T_1]/(P_1)$. Then we have an exact sequence

$$0 \rightarrow P_1 k[T_1, \dots, T_n] \rightarrow k[T_1, \dots, T_n] \rightarrow k_1[T_2, \dots, T_n] \rightarrow 0.$$

- (b) There exist n polynomials $P_1(T_1), P_2(T_1, T_2), \dots, P_n(T_1, \dots, T_n)$ such that $k[T_1, \dots, T_i] \cap \mathfrak{m} = (P_1, \dots, P_i)$ for all $i \leq n$. In particular, $\mathfrak{m}$ is generated by n elements.

Proof. Part (a) is obvious. In part (b), we proceed by induction. The statement is obvious in the single-variable case, so suppose that it holds for $n - 1$, with n an integer at least 2. Let k_1 be as in (a) and observe that it is a field, so the hypotheses and the exact sequence from (a) give a generating set $(P_2, \dots, P_n)$ for the image of $\mathfrak{m}$ in $k_1[T_2, \dots, T_n]$. This pulls back to a set of generators $(P_1, \dots, P_n)$ for $\mathfrak{m}$, and it is easy to see that the desired properties hold. $\square$

Exercise 2.1.6. Any closed subspace of a quasicompact space is itself quasicompact. If A is a ring, $\text{Spec}(A)$ is quasicompact.

Proof. Let X be a topological space and let $Y \subseteq X$ be any closed subset. If $\{U_i\}$ is an open covering of Y , $\{U_i \cup \complement_X Y\}$ is an open covering of X . Taking a finite subcovering of this extended covering yields a finite subcovering of the original covering.

For the second statement, suppose that $\{U_i\}$ is an open covering of $X = \operatorname{Spec} A$. For each i , let J_i be the ideal such that $\complement_X U_i = V(J_i)$. We see that $\emptyset = \bigcap V(J_i) = V(\sum J_i)$, so there is a finite family of $f_k \in J_k$ such that $\sum f_k = 1$. Then $\sum J_k = 1$ and $\{U_k\}$ is a finite subcovering. $\square$

The next exercise is a bit open ended in character, so the proof label is changed to reflect that.

Exercise 2.1.7. Describe the closed subsets of $\operatorname{Spec}(\mathbf{Z}[T])$. Show that they are not all principal.

Answer. The closed subsets of $\operatorname{Spec}(\mathbf{Z}[T])$ can be broken up into three families: points, lines, and the generic point. The points are simply maximal ideals; in $\mathbf{Z}[T]$ these are ideals of the form (p, P) , where p is a non-zero prime number and P is an irreducible polynomial. The lines are just principal ideals; I call them lines because they determine a family of points that looks kind of like a line. Finally, there is the generic point (0) . The fact that the closed subsets are not all principal follows from the fact that the maximal ideals have 2 generators. $\square$

Exercise 2.1.8. Let $\varphi: A \rightarrow B$ be an integral homomorphism.

- (a) The map $\operatorname{Spec} \varphi: \operatorname{Spec} B \rightarrow \operatorname{Spec} A$ sends closed points to closed points, and any point in the preimage of a closed point is also a closed point.
- (b) Let $\mathfrak{p} \in \operatorname{Spec} A$. The canonical homomorphism $A_{\mathfrak{p}} \rightarrow B \otimes_A A_{\mathfrak{p}}$ is integral.
- (c) Let $T = \varphi(A \setminus \mathfrak{p})$ and suppose that φ is injective. Then T is a multiplicative subset of B and $B \otimes_A A_{\mathfrak{p}} = T^{-1}B \neq 0$. In particular, $\operatorname{Spec} \varphi$ is surjective if φ is integral and injective.

Proof. For part (a), suppose that $\mathfrak{m}$ is a closed point of $\operatorname{Spec} B$ and let $\varphi^{-1}(\mathfrak{m}) = \mathfrak{m}'$. Then we have an integral homomorphism $A/\mathfrak{m}' \hookrightarrow B/\mathfrak{m}$. Given $a \in A/\mathfrak{m}'$, the inverse $b \in B/\mathfrak{m}$ of its image under φ satisfies an integral equation $a_0 + a_1 b + \dots + b^n = 0$. Multiplying by an appropriate power of a , we have $a_0 a^n + \dots + 1 = 0$, so a is a unit. On the other hand, let $\mathfrak{m}$ be a closed point of $\operatorname{Spec} A$ and suppose that $\varphi(\mathfrak{m}) \subseteq \mathfrak{m}'$ for some $\mathfrak{m}' \in \operatorname{Spec} B$. Again, we have an integral homomorphism $A/\mathfrak{m} \hookrightarrow B/\mathfrak{m}'$. Since $A/\mathfrak{m}$ is a field, this is an algebraic extension.

Parts (b) and (c) are easy, but somewhat annoying to write out. $\square$

Exercise 2.1.9. Let k be a field and A be a finitely generated k -algebra.

- (a) Suppose in addition that A is a finite over k . Then $\operatorname{Spec} A$ is a finite set, with cardinality bounded above by $\dim_k A$. Furthermore, every prime ideal of A is maximal.

(b) The cardinality of $\text{Spec}(k[T_1, \dots, T_d])$ is infinite when $d \geq 1$.

(c) The set $\text{Spec } A$ is finite iff A is finite over k .

Proof. For part (a), the second statement is obvious. To prove the first one, let $\{a_1, \dots, a_n\}$ be a basis for A over k . We see that there is a family of maximal ideals $\mathfrak{m}_1, \dots, \mathfrak{m}_n$, where $\mathfrak{m}_i$ is the largest ideal not containing a_i . If $\mathfrak{m}$ is any other maximal ideal, there is some i such that $a_i \notin \mathfrak{m}$, so $\mathfrak{m} = \mathfrak{m}_i$.

In (b), it suffices to prove it when $d = 1$. In this case, it's just Euclid's proof of the infinitude of primes.

Part (c) follows from parts (a) and (b) after an easy application of Noether normalization. $\square$

2.2 Ringed topological spaces

Let X be a topological space.

Exercise 2.2.1. Let A be a non-trivial abelian group and let $\mathcal{A}_X$ be the constant presheaf of A on X . The associated sheaf $\mathcal{A}_X^\sharp$ is the sheaf of locally constant functions valued in A . In particular, $\mathcal{A}_X$ is a sheaf if and only if every non-empty open subset of X is connected.

Proof. Let $\mathcal{F}$ be the sheaf of locally constant A -valued functions on X and let $\varphi: \mathcal{A}_X \rightarrow \mathcal{F}$ be the evident morphism. It is easy to see that the stalks of $\mathcal{F}$ are all A . Let $\mathcal{G}$ be any sheaf and let $\alpha: \mathcal{A}_X \rightarrow \mathcal{G}$ be any morphism. If f is a locally constant function on U , we may split U into disjoint open sets $U_a = f^{-1}(a)$ for $a \in f(U)$. We define $\tilde{\alpha}(U): \mathcal{F} \rightarrow \mathcal{G}$ to be the map sending f to the regluing of the sections $\alpha(U)(a)|_{U_a}$. It is easy to see that this defines a morphism of sheaves and that $\tilde{\alpha} \circ \varphi = \alpha$. Since φ is surjective, $\tilde{\alpha}$ is the unique such morphism. $\square$

Exercise 2.2.2. Let $\mathcal{F}$ be a sheaf on X . Let $s, t \in \mathcal{F}(X)$. Then the set of $x \in X$ such that $s_x = t_x$ is open in X .

Proof. If $s_x = t_x$ at some point $x \in X$, there must be a neighborhood U_x of x such that $s_y = t_y$ for all $y \in U_x$. Taking the union of all such U_x gives the result. $\square$

Exercise 2.2.3. Let X be a fixed topological space. Let $\mathcal{F}$ be a presheaf on X . We keep the notation introduced just before Lemma 2.7.

- (a) Let $\mathcal{U} = \{U_i\}$ be an open covering of X . Let $\mathcal{F}_{\mathcal{U}}(X) = \text{Ker } d_1$. For any open subset W of X , we define in the same way a group $\mathcal{F}_{\mathcal{U}}(W)$ relative to the covering $\{W \cap U_i\}$ of W . Then $\mathcal{F}_{\mathcal{U}}$ is a presheaf on X and we have a morphism of presheaves $\mathcal{F} \rightarrow \mathcal{F}_{\mathcal{U}}$.
- (b) Let $\mathcal{V} = \{V_j\}$ be another open covering of X . We define a partial ordering: $\mathcal{V} \preceq \mathcal{U}$ if $\mathcal{V}$ is finer than $\mathcal{U}$. Then there is a canonical homomorphism $\mathcal{F}_{\mathcal{U}}(W) \rightarrow \mathcal{F}_{\mathcal{V}}(W)$ for every open subset W if $\mathcal{V} \preceq \mathcal{U}$.

- (c) The ordering $\preceq$ makes the set of open coverings of X into a direct system. Let

$$\mathcal{F}'(W) = \varinjlim_{\mathcal{V}} \mathcal{F}_{\mathcal{V}}(W),$$

the direct limit being taken over the open coverings of W . Then $(\mathcal{F}')'$ is the sheaf associated to $\mathcal{F}$.

Proof. Part (a) is clear. The homomorphisms in part (b) are given by the restrictions in the natural way.

Part (c) is quite nasty to write out so I won't do it. $\square$

Exercise 2.2.4. Let $0 \rightarrow \mathcal{F}' \rightarrow \mathcal{F} \rightarrow \mathcal{F}''$ be an exact sequence of sheaves on X . Then the sequence of abelian groups $0 \rightarrow \mathcal{F}'(X) \rightarrow \mathcal{F}(X) \rightarrow \mathcal{F}''(X)$ is exact.

Proof. This follows from the exact sequence induced on stalks. $\square$

Exercise 2.2.5. Fix a sheaf $\mathcal{G}$ on X and a closed point $x_0 \in X$. Define a presheaf $\mathcal{F}$ by $\mathcal{F}(U) = \mathcal{G}(U)$ if $x_0 \notin U$ and $\mathcal{F}(U) = \{s \in \mathcal{G}(U) : s_{x_0} = 0\}$ otherwise. Then $\mathcal{F}$ is a sheaf and $\text{Supp } \mathcal{F} = \text{Supp } \mathcal{G} \setminus \{x_0\}$.

Proof. I think I'm missing something because this seems really immediate $\square$

Exercise 2.2.6. Let $f: X \rightarrow Y, g: Y \rightarrow Z$ be continuous maps, $\mathcal{F}$ a sheaf on X and $\mathcal{H}$ a sheaf on Z . Then

$$f^{-1}(g^{-1}\mathcal{H}) = (g \circ f)^{-1}\mathcal{H}, g_*(f_*\mathcal{F}) = (g \circ f)_*\mathcal{F}.$$

Proof. The second part is obvious. For the first one, notice that the assignment

$$\mathcal{F} \mapsto \left(U \mapsto \varprojlim_{V \supseteq f(U)} \mathcal{F}(U) \right)$$

is functorial, so we have a morphism $f^{-1}(g^{-1}\mathcal{H}) \rightarrow (g \circ f)^{-1}\mathcal{H}$. It is easy to check that this restricts to isomorphisms on the stalks, so it is an isomorphism. We can say this is equality because the isomorphism is sufficiently special. $\square$

Exercise 2.2.7. Let $\mathcal{B}$ be a base of open subsets for a topological space X . Let $\mathcal{F}, \mathcal{G}$ be two sheaves on X . Suppose that for every $U \in \mathcal{B}$, there is a homomorphism $\alpha(U): \mathcal{F}(U) \rightarrow \mathcal{G}(U)$ which is compatible with restrictions. Then there is a unique extension $\alpha: \mathcal{F} \rightarrow \mathcal{G}$ to a homomorphism of sheaves. Moreover, if $\alpha(U)$ is surjective (resp. injective) for every $U \in \mathcal{B}$, then α is surjective (resp. injective).

Proof. Let U be any open set and s a section of $\mathcal{F}$ over U . If $\{U_i\}$ is a cover of U by elements of $\mathcal{B}$, we have a corresponding family $\{s_i\}$ of sections of $\mathcal{F}$ given by restrictions of s . Since α is compatible with restrictions, we obtain a family $\{\alpha(U_i)(s_i)\}$ that agrees on intersections, which reglues into a section $\alpha(s)$ of $\mathcal{G}$. It is easy to see that this is a morphism of sheaves, and the uniqueness of this extension follows from the uniqueness of gluing. The statement about surjectivity and injectivity is obvious. $\square$

Exercise 2.2.8. Let $\{U_i\}$ be an open covering of X . Suppose we are given a family of sheaves $\mathcal{F}_i$ on U_i and isomorphisms $\phi_{ij}: \mathcal{F}_i|_{U_i \cap U_j} \rightarrow \mathcal{F}_j|_{U_i \cap U_j}$ such that $\phi_{ik} = \phi_{jk} \circ \phi_{ij}$ on $U_i \cap U_j \cap U_k$ and $\phi_{ii} = 1_{\mathcal{F}_i}$. Then there is a unique sheaf $\mathcal{F}$ on X with isomorphisms $\psi_i: \mathcal{F}|_{U_i} \rightarrow \mathcal{F}_i$ such that $\psi_j = \phi_{ij} \circ \psi_i$ on $U_i \cap U_j$.

Proof. The sheaf $\mathcal{F}$ is quite simple in its definition. For any open subset V of X , we define a section of $\mathcal{F}(V)$ to be a family $\{s_i\}$ such that $s_i \in \mathcal{F}_i(V \cap U_i)$ and $s_i|_{V \cap U_i \cap U_j} = \phi_{ji}(s_j|_{V \cap U_i \cap U_j})$ for all i, j . It is easy to see that this defines a sheaf, and the existence of the desired isomorphisms is clear. To see that it is unique, let $\mathcal{G}$ be another such sheaf, and denote the isomorphisms $\mathcal{G}|_{U_i} \rightarrow \mathcal{F}_i$ by ψ'_i . Then we have a family of isomorphisms $\mathcal{G}|_{U_i} \rightarrow \mathcal{F}|_{U_i}$, which we may glue to an isomorphism $\mathcal{G} \rightarrow \mathcal{F}$. $\square$

Exercise 2.2.9. Let $\mathcal{F}$ be a sheaf on X whose support Y is a finite set of closed points y . Then for any open set U of X , $\mathcal{F} = \bigoplus_{x \in U \cap Y} \mathcal{F}_x$.

Proof. Let U be an open subset of X meeting Y at only a single point y . Then for any open neighborhood V of y contained in U , we have $U = (U \setminus \{y\}) \cup V$. Since $\mathcal{F}(U \setminus \{y\}) = 0$, the value of any section $s \in \mathcal{F}(U)$ is determined by its restriction to V , hence by its stalk at y . The statement now follows from an easy gluing argument. $\square$

2.3 Schemes

Exercise 2.3.1. Let X be an open subscheme of an affine scheme Y . The canonical morphism $X \rightarrow Y$ corresponds by the map ρ of Proposition 25 to the restriction $\mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(X)$.

Proof. In the case where X is a principal open subset, this is just Lemma 3.7. Otherwise, suppose that $X = \bigcup D(f_i)$ for some family of elements $f_i \in \mathcal{O}_Y(Y)$. The inclusions induce a family of maps $\mathcal{O}_Y(Y) \rightarrow \mathcal{O}_X(D(f_i)) = \mathcal{O}_X(X)_{f_i}$ which clearly agree on the intersections. Gluing these gives the restriction. $\square$

Exercise 2.3.2. Let $U = \text{Spec } B$ be an affine open subscheme of $X = \text{Spec } A$. Then the restriction $A \rightarrow B$ is flat.

Proof. Since the restriction corresponds to the open immersion $U \hookrightarrow X$, we see that its localization $A_x \rightarrow B_x$ is an isomorphism, which is flat, for every $x \in \text{Spec } B$. $\square$

Exercise 2.3.3.

- (a) Let F be a subset of an affine scheme $\text{Spec } A$. The closure $\overline{F}$ of F in $\text{Spec } A$ is equal to $V(I)$, where $I = \bigcap_{\mathfrak{p} \in F} \mathfrak{p}$.
- (b) Let $\varphi: A \rightarrow B$ be a ring homomorphism. Let $f: \text{Spec } B \rightarrow \text{Spec } A$ be the morphism of schemes associated to φ . Then $\overline{\text{Im } f} = V(\text{Ker } \varphi)$.

Proof. We already have that $\overline{F}$ must already be of the form $V(I')$ with $I' \subset I$. Since $V(I')$ must be the minimal such set, we have $I' = I$.

For part (b) we assume that A and B are reduced, since that doesn't affect their prime spectra. Then the previous part implies that $\overline{\text{Im } f} = V(\bigcap_{\mathfrak{p} \in \text{Spec } A} \varphi^{-1}(\mathfrak{p})) = V(\text{Ker } \varphi)$. $\square$

In the case where $B = A_{\mathfrak{p}}$ for some $\mathfrak{p} \in \text{Spec } A$, this is the set of ideals that are "large enough" to interact with $\mathfrak{p}$. In geometric terms, they are the irreducible varieties such that every function that vanishes on $V(\mathfrak{p})$ vanishes here as well.

Exercise 2.3.4. Let X be a scheme and $f \in \mathcal{O}_X(X)$. Then the map $U \mapsto f|_U \mathcal{O}_X(U)$ for every affine open subset U defines a sheaf of ideals on X , denoted $f\mathcal{O}_X$. In addition, $\text{Supp } f\mathcal{O}_X$ is closed.

Proof. It is easy to see that $f\mathcal{O}_X$ inherits the uniqueness condition from $\mathcal{O}_X$. On the other hand, suppose that we have an open set U of X , an affine open cover $\{U_i\}$ of U , and a family of sections $f|_{U_i} s_i$ that agrees on intersections. Then we may glue these into a section s of $\mathcal{O}_X(U)$ that restricts to $f|_{U_i} s_i$ on every U_i , which we contest is contained in $f\mathcal{O}_X(U)$. Indeed, suppose that s is not in this ideal. Then for any $a \in \mathcal{O}_X(U)$, there is some $x \in U$ such that $s_x - (af)_x \neq 0$, but this contradicts the fact that $f|_{U_i} s_i$ is in $f\mathcal{O}_X(U)$.

To see that $\text{Supp } f\mathcal{O}_X$ is closed, observe that $\text{Supp } f\mathcal{O}_X$ is just the set of $x \in X$ such that $f_x \neq 0$. Since the complement is open, the statement holds. $\square$

Exercise 2.3.5. Let Y be a scheme such that $\text{Mor}(X, Y) \cong \text{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_X(X))$ for any affine scheme X . Then Y is affine.

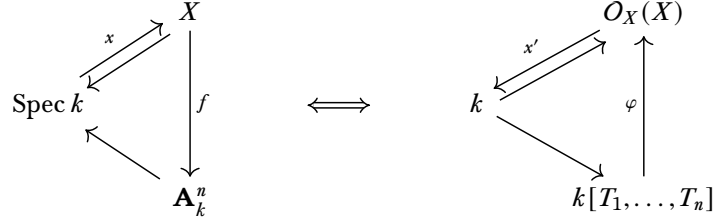
Proof. Regrettably, the easiest proof I could come up with was by category slop. Observe that for any affine scheme Z , we have $\text{Mor}(Z, Y) \cong \text{Hom}(\mathcal{O}_Y(Y), \mathcal{O}_Z(Z)) \cong \text{Mor}(Z, \text{Spec}(\mathcal{O}_Y(Y)))$. This extends to the category of all schemes in the obvious way, so Y is a representing object of $\text{Mor}(-, \text{Spec}(\mathcal{O}_Y(Y)))$ and is thus isomorphic to $\text{Spec}(\mathcal{O}_Y(Y))$. $\square$

Exercise 2.3.6. Let A be a ring and consider the A -scheme $X = \text{Spec}(A[T_1, \dots, T_n]/I)$. The A -points of X are in bijective correspondence with the n -tuples $\alpha = (\alpha_1, \dots, \alpha_n) \in A^n$ such that $P(\alpha) = 0$ for all $P \in I$.

Proof. An A -point of X is the same as a homomorphism $A[T_1, \dots, T_n]/I \rightarrow A$ that restricts to the identity on A , so it is determined only by the values of the T_i . This homomorphism must send $T = (T_1, \dots, T_n)$ to a family $\alpha = (\alpha_1, \dots, \alpha_n)$ satisfying the condition above. The statement is now obvious. $\square$

Exercise 2.3.7. Let X be a scheme over the field k . Let $\varphi: k[T_1, \dots, T_n] \rightarrow \mathcal{O}_X(X)$ be a homomorphism of k -algebras and $f: X \rightarrow \mathbf{A}_k^n$ the morphism induced by φ . Then for any rational point $x \in X(k)$, via the identification $\mathbf{A}_k^n = k^n$, we have $f(x) = (f_1(x), \dots, f_n(x))$, where $f_i = \varphi(T_i)$ and $f_i(x)$ is the image of f_i in $k(x) = k$.

Proof. The statement is obvious if you draw a picture and use the universal property of affine schemes:



□

Exercise 2.3.8. Let X be a scheme over a ring A . Let $f_0, \dots, f_n \in O_X(X)$ be such that the $f_{i,x}$ generate the unit ideal of $O_{X,x}$ for all $x \in X$. Then X is the union of the X_{f_i} . Moreover, there is a morphism $f: X \rightarrow \mathbf{P}_A^n$ such that $f^{-1}(D_+(T_i)) = X_{f_i}$ and $f|_{X_{f_i}}$ is induced by the homomorphism $A[T_i^{-1}T_j]_j \rightarrow O_X(X_{f_i})$ given by $T_i^{-1}T_j \mapsto f_i^{-1}f_j$.

Proof. Let $x \in X$ be any point. Since the $f_{i,x}$ generate the unit ideal, there is j such that $f_{j,x}$ is not contained in the maximal ideal at x , so $x \in X_{f_j}$. On the other hand, we have a homomorphism $A[T_0, \dots, T_n] \rightarrow O_X(X)$ given by $T_i \mapsto f_i$. This gives rise to a homomorphism $A[T_i^{-1}T_j]_j \rightarrow O_X(X)_{f_i} \cong O_X(X_{f_i})$ for any i , hence a morphism $X_{f_i} \rightarrow D_+(T_i)$. We contest that this agrees on intersections. Indeed, observe that for any i, j , the morphisms $X_{f_i} \rightarrow D_+(T_i)$ and $X_{f_j} \rightarrow D_+(T_j)$, when restricted to $X_{f_i f_j} = X_{f_i} \cap X_{f_j}$, both factor through $D_+(T_i) \cap D_+(T_j) = D_+(T_i T_j)$; it is then clear that they agree, so we have the desired morphism $X \rightarrow \mathbf{P}_A^n$. □

Note: I omitted the last part of 2.3.8 because it is the same as 2.3.7.

Exercise 2.3.9. Let B be a graded ring. Let $f \in B$ be a non-nilpotent, homogeneous element of degree 0. Then B_f possesses a natural grading, and $D_+(f) = \text{Proj } B_f$.

Proof. The grading is obvious. Recall that a prime ideal only localizes to a prime in B_f if it does not contain f . Then $D_+(f)$ is just $\text{Proj } B_f$. □

2.4 Reduced schemes

Exercise 2.4.1. Let k be a field and $P \in k[T_1, \dots, T_n]$. Then $\text{Spec}(k[T_1, \dots, T_n]/(P))$ is a reduced (resp. irreducible; integral) if and only if P is squarefree (resp. admits only one irreducible factor; is irreducible).

Proof. dawg □

Exercise 2.4.2. Let X be a scheme and $x \in X$. Then the image of the morphism $\text{Spec } O_{X,x} \rightarrow X$ is the set of points of X that specialize to x .

Proof. It suffices to prove this in the case where X is affine. If y is any point of X that specializes to x , we have $\mathfrak{j}_y \subseteq \mathfrak{j}_x$ and so y is in the image of $\text{Spec } \mathcal{O}_{X,x}$. The other direction is basically the same. $\square$

Exercise 2.4.3. Let $\mathcal{O}_K$ be a discrete valuation ring with field of fractions K and uniformizing parameter t . Then $\text{Spec } K[T]$ embeds as an open subscheme of $\text{Spec } \mathcal{O}_K[T]$.

Proof. Observe that $(\mathcal{O}_K)_t = K$, so $(\mathcal{O}_K[T])_t = K[T]$. Since $\text{Spec}((\mathcal{O}_K[T])_t)$ is homeomorphic to $D(t) \subseteq \text{Spec}(\mathcal{O}_K[T])$, this proves the proposition. $\square$

The second part of this exercise is to find the points of $\text{Spec } K[T]$ which specialize to the point corresponding to the maximal ideal (T, t) of $\mathcal{O}_K[T]$. To do this, notice that $\text{Spec } K[T]$ corresponds to the ideals I of $\mathcal{O}_K[T]$ such that $I \cap (t) = \{0\}$, i.e. the principal ideals generated by nonconstant polynomials. Then the only points that specialize to (T, t) are (0) and (T) .

Exercise 2.4.4. Let X be a scheme having only a finite number of irreducible components $\{X_i\}$. Then X is connected if and only if for any pair i, j , there exist indices $i_0 = i, i_1, \dots, i_r = j$ such that $X_{i_l} = X_{i_{l+1}}$ for every $l < r$. Furthermore, X is integral if and only if it is connected and integral at every point.

Proof. Since the irreducible components of X are finite in number, they must all be open; it is clear that having X connected implies the condition. On the other hand, suppose that X has the property. If U is a nonvoid open set such that $\overline{U} = U$, we have that U is the union of irreducible components of X , so suppose that X_i is any component (not necessarily contained in U). Then we have a sequence $X = X_0, \dots, X_r \subseteq U$ such that $X_i \cap X_{i+1} \neq \emptyset$. However, since $X_{r-1} \cap U \neq \emptyset$, X_{r-1} is contained in U and so we may drop the last index. Repeating this, we may assume that $r = 0$ and so $X \subseteq U$.

Now, if X is integral, it is obvious that it is connected and integral at every point, so assume the latter. At any point $x \in X$ let $i: \text{Spec } \mathcal{O}_{X,x} \rightarrow X$ be the canonical morphism. Since $i^{-1}(i((0))) \subseteq i^{-1}(\overline{i((0))})$, we have that $\text{Spec } \mathcal{O}_{X,x} \subseteq i^{-1}(\overline{i((0))})$ and so the image of i has a closed point. It follows from exercise 2 that each point x is contained in exactly one irreducible component, so the connectedness hypothesis implies that X is irreducible. That X is reduced is obvious, so X is integral. $\square$

Exercise 2.4.5. Let X be a scheme. Every irreducible component of X is contained in a connected component. If X is locally noetherian, then the connected components are open. If X is noetherian, the connected components are finite in number.

Proof. The first and third statements are clear, and the second one follows from the third. $\square$

Exercise 2.4.6. Let A be a ring and X be a scheme.

- (a) The following properties are equivalent:
- (i) X is connected;
 - (ii) $\mathcal{O}_X(X)$ has no idempotent elements other than 1;
 - (iii) $\text{Spec}(\mathcal{O}_X(X))$ is connected.
- (b) Any local scheme is connected.
- (c) Suppose that the connected components of X are open. Let U be a connected component of X . Then there is a unique idempotent element e of $\mathcal{O}_X(X)$ such that $e|_U = 1$ and $e|_{X \setminus U} = 0$. This establishes a canonical bijection from the set of connected components of X onto the set of indecomposable idempotent elements of $\mathcal{O}_X(X)$, the inverse map being given by $e \mapsto V(1 - e)$.

Proof. For part (a), let $e \neq 1 \in \mathcal{O}_X(X)$ be an idempotent element. At any point $x \in X$, e_x must also be idempotent, so generates an ideal (e_x) that squares to itself. If $e_x \neq 1$, Nakayama's lemma implies that it is 0. We see that the sets $U = \{x \in X: e_x = 1\}$ and $V = \{x \in X: e_x = 0\}$ are disjoint and form an open cover of X , so (i) implies (ii).

On the other hand, suppose that U, V are disjoint open sets that cover $\text{Spec}(\mathcal{O}_X(X))$. Then there is an element of $\mathcal{O}_X(X)$ that restricts to 0 on U and 1 on V . It is clearly idempotent and not equal to 1, so (ii) implies (iii).

Finally, suppose that X is disconnected. Then the same argument as in part (ii) implies that there is an idempotent $e \neq 1 \in \mathcal{O}_X(X)$, hence $\text{Spec}(\mathcal{O}_X(X))$ is disconnected and so (a) follows.

Part (b) follows from the same reasoning as part (a)(ii), and part (c) is obvious. $\square$

Exercise 2.4.7. The underlying topological space of a scheme is a Kolmogorov space.

Proof. It suffices to show the statement for affine schemes, so let X be an affine scheme and let x and y be distinct points of X . If $\mathfrak{j}_x \subset \mathfrak{j}_y$, $X \setminus V(\mathfrak{j}_y)$ is a neighborhood of x not containing y . Otherwise, suppose there is no inclusion relation between $\mathfrak{j}_x$ and $\mathfrak{j}_y$. Then there is some $f \in \mathfrak{j}_x$ such that $f \notin \mathfrak{j}_y$, so $D(f)$ is a neighborhood of y not containing x . $\square$

Exercise 2.4.8. A quasicompact scheme has a closed point.

Proof. Let X be a scheme with a finite cover $\{U_i\}$ by affine opens (assuming no inclusion relation between the U_i) and suppose that $x \in U_1$ is a closed point in U_1 (but not necessarily in X !). If x is a closed point in X , we are done; otherwise, there is another affine open in the cover, say U_2 , such that $\overline{\{x\}}$ meets U_2 at a point other than x . It is clear that $\overline{\{x\}}$ contains a closed point of U_2 which is not contained in U_1 . Continuing this process will give the desired point. $\square$

Exercise 2.4.9. Let X be a noetherian scheme. Then the set of points such that $\mathcal{O}_{X,x}$ is reduced (resp. is an integral domain) is open.

Proof. First, assume X is the prime spectrum of a noetherian ring. Denote by N the nilradical of $\mathcal{O}_X(X)$, and let $f_1, \dots, f_n$ be generators of N . Then for any $x \in X$, $\mathcal{O}_{X,x}$ is reduced only if $\text{Ann}(f_i)$ is not contained in $\mathfrak{j}_x$ for all i , i.e. only if $x \in D(f_1 f_2 \dots f_n)$. An easy gluing argument gives the statement for general X . $\square$

3 Morphisms and base change

3.1 The technique of base change

Exercise 3.1.1. Let S , X , and Y be sets, and equip X (resp. Y) with a map $\varphi: X \rightarrow S$ (resp. $\psi: Y \rightarrow S$). Let $p: X \times_S Y \rightarrow X$ and $q: X \times_S Y \rightarrow Y$ be the restrictions of the projections from $X \times Y$. Then for any set Z , equipped with a pair of maps $f: Z \rightarrow X$ and $g: Z \rightarrow Y$ such that $\varphi \circ f = \psi \circ g$, there is a unique map $(f, g)_S: Z \rightarrow X \times_S Y$ such that $p \circ (f, g)_S = f$ and $q \circ (f, g)_S = g$.

Proof. The existence (but perhaps not the uniqueness) of $(f, g)_S$ is clear, so suppose that $h: Z \rightarrow X \times_S Y$ is such that $p \circ h = f$ and $q \circ h = g$. Then for all $z \in Z$, $h(z) = (p(h(z)), q(h(z))) = (f(z), g(z)) = (f, g)_S(z)$ and so $h = (f, g)_S$. $\square$

Exercise 3.1.2. Let $f: X \rightarrow Y$ be a morphism of schemes. For any scheme T , let $f(T): X(T) \rightarrow Y(T)$ denote the map defined by $f(T)(g) = f \circ g$. Then $f(T)$ is bijective for every T if and only if f is an isomorphism.

Proof. Yoneda $\square$

Exercise 3.1.3. Let $f: X \rightarrow Y$ be a dominant morphism of irreducible schemes. Then the generic fiber of f is dense in X .

Proof. Let ζ_X and ζ_Y be the generic points of X and Y , respectively. Then Proposition 1.16 implies that ζ_X is in the generic fiber, so it is dense. $\square$

Exercise 3.1.4. Let $f: X \rightarrow Y$ be a morphism of affine schemes. Then for any affine open subscheme V of Y , $f^{-1}(V)$ is affine.

Proof. Just recall that $f^{-1}(V) = X \times_Y V$, so is affine. $\square$